

Multiplying by a One-Digit Number

Partial products, the standard algorithm, and estimating before you multiply

Directions:

- Show your work in the space under each problem — the method matters as much as the answer.
- In the standard algorithm, write each regrouped digit above the next column before you use it.
- Where a problem asks for partial products, fill every blank, then add them for the total.

1. Multiply. 63×4

Answer: _____

2. Multiply. 58×7

Answer: _____

3. Break 214 into its place values and multiply each part by 3:

$$3 \times 200 = \underline{\hspace{2cm}} \quad 3 \times 10 = \underline{\hspace{2cm}} \quad 3 \times 4 = \underline{\hspace{2cm}}$$

Now add the three partial products to get 214×3 .

Answer: _____

4. Complete the area model for $1,325 \times 4$, then add the parts.

$\times$	1,000	300	20	5
4	_____	_____	_____	_____

Total:

Answer: _____

5. Multiply. $3,067 \times 5$

Answer: _____

6. Estimate first, then multiply exactly: 412×6

(a) Round 412 to the nearest hundred. _____

(b) Use your rounded number to estimate the product. _____

(c) Now find the exact product. _____

7. Multiply. $2,415 \times 8$

Answer: _____

8. A theatre has 1,248 seats. It is sold out for 7 shows in a row.

(a) Exactly how many seats are sold across the 7 shows?

Answer: _____ seats

(b) Round that total to the nearest thousand. _____

9. Multiply. $5,704 \times 9$

Answer: _____

Answer: _____

Answer: _____ bricks

(b) The tens digit of 1,608 is 0, but the tens digit of the product is not 0. Explain why.